```
# Install libraries
!pip install pandas numpy scikit-learn matplotlib

# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error

# ──────────────────────────────
# 1. Create Dataset
# ──────────────────────────────
data = pd.DataFrame({
    'Study_Hours': [1,2,3,4,5,6,7,8,9,10],
    'Attendance': [50,55,60,65,70,75,80,85,90,95],
    'Previous_Marks': [40,45,50,55,60,65,70,75,80,85],
    'Final_Marks': [42,48,52,58,63,67,72,78,84,90]
})

print("Dataset:\n")
print(data)

# ──────────────────────────────
# 2. Preprocessing
# ──────────────────────────────
print("\nMissing Values:\n", data.isnull().sum())

# ──────────────────────────────
# 3. Feature Extraction
# ──────────────────────────────
X = data[['Study_Hours', 'Attendance', 'Previous_Marks']]
y = data['Final_Marks']

# ──────────────────────────────
# 4. Training
# ──────────────────────────────
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

# ──────────────────────────────
# 5. Testing
# ──────────────────────────────
predictions = model.predict(X_test)

results = pd.DataFrame({
    'Actual': y_test.values,
    'Predicted': predictions
})

print("\nResults:\n")
print(results)

# ──────────────────────────────
# 6. Evaluation
# ──────────────────────────────
rmse = np.sqrt(mean_squared_error(y_test, predictions))
print("\nRMSE:", rmse)

# ──────────────────────────────
# 7. Visualization
# ──────────────────────────────
plt.figure()
plt.scatter(y_test, predictions)
plt.xlabel("Actual Marks")
plt.ylabel("Predicted Marks")
plt.title("Student Performance Prediction")
plt.show()
```

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# ────────────────────────────
# 8. Predict New Data (Extra)
# ────────────────────────────
new_data = [[8, 85, 75]]  # Study Hours, Attendance, Previous Marks
predicted_mark = model.predict(new_data)

print("\nPredicted Mark for new student:", predicted_mark[0])
```

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Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0
Dataset:

   Study_Hours  Attendance  Previous_Marks  Final_Marks
0            1          50              40           42
1            2          55              45           48
2            3          60              50           52
3            4          65              55           58
```